

SURVEY

LiDAR-Based Obstacle Avoidance With Autonomous Vehicles: A Comprehensive Review

PUI YEE LEONG^{ID} AND NUR SYAZREEN AHMAD^{ID}, (Member, IEEE)

School of Electrical and Electronic Engineering, Universiti Sains Malaysia, Nibong Tebal, Penang 14300, Malaysia

Corresponding author: Nur Syazreen Ahmad (syazreen@usm.my)

This work was supported in part by Universiti Sains Malaysia (USM) under USM-Industry Matching Grant Scheme under Project 2023/0254/EM0023, and in part by the Industrial Grant from SanDisk Storage Malaysia Sdn. Bhd. under Project 2023/0033/AO2720.

ABSTRACT Research on autonomous vehicles (AV) - self-navigating machines that transport both man and cargo has proliferated lately. While once limited to industrial or military uses, more attention is now given to their potential applications in broader society, especially in taking over the mundane, risky and taxing jobs from humans. From self-driving cars, to autonomous mobile robots, to unmanned air, surface and underwater vehicles, one challenge common to all of them is the need to navigate autonomously without human intervention as they operate in their intended environment. The ability to accurately detect and safely avoid obstacles is thus imperative to achieving greater autonomy in vehicles. In recent years, light detection and ranging (LiDAR) sensors - known for their accuracy and reliability in measuring distances- have been widely used for obstacle avoidance. However, as AVs are expected to function under a multitude of conditions, the usage of a single sensor is insufficient. Sensor fusion becomes the next logical step to allow the vehicle to detect and respond to a wider variety of situations. In this paper, we investigate the ways sensor fusion can be applied to improve the obstacle avoidance capability of various indoor and outdoor LiDAR-based AVs by reviewing recent publications over the past decade. The core contribution includes examining the types of secondary sensor used, the motivation behind their selection, as well as the obstacle avoidance algorithm used. Finally, the obstacle avoidance research trends driving indoor and outdoor AVs are discussed and future research directions are presented.

INDEX TERMS Autonomous vehicles, autonomous mobile robots, LiDAR, obstacle avoidance, sensor fusion.

I. INTRODUCTION

Throughout the course of civilization, humans have sought the automation of labor; from ancient wind or water-powered mills, to the steam and electric engines of more modern times, it is the nature of man to relegate the mundane, risky and taxing jobs to machines. Recent technological advancements have now allowed us to achieve varying degrees of autonomy in machines, where work can be continuously produced under various operating circumstances with minimal human instruction. It comes as no surprise then, as vehicles, which has long served to move loads - both people and cargo alike,

are among the most researched for transition towards full autonomy [1].

Unmanned or autonomous ground, sea and air vehicles, while having its roots in military or manufacturing domains, have now captured the attention of the greater public with promises of further convenience, safety and efficiency [2]. Self-driving or autonomous cars (AC) can potentially free drivers from the stressful daily commute. Autonomous mobile robots (AMR) serve a wide range of purposes across multiple sectors, from warehouse logistics to providing services to human customers. Drones or unmanned aerial vehicles (UAV), as well as unmanned surface vehicles (USV), are poised to take the skies and seas respectively, aiding in research, environmental monitoring and also the transportation and delivery of cargo. While not the focus of

The associate editor coordinating the review of this manuscript and approving it for publication was Md. Abdur Razzaque^{ID}.

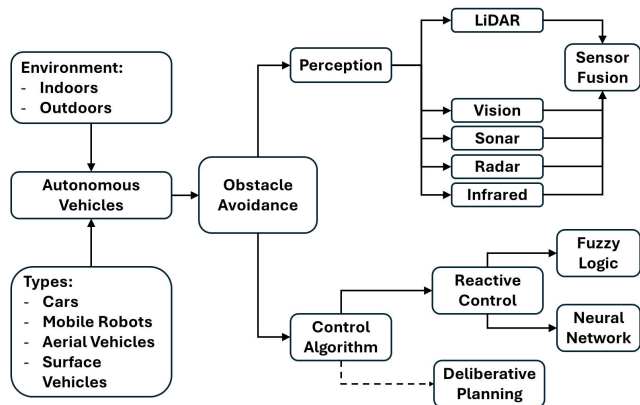


FIGURE 1. Overview of the framework of this study: The main aspects of obstacle avoidance explored in this work are marked with solid arrows.

this work, autonomous undersea vehicles are being deployed for seafloor exploration and rescue missions, as autonomous space vehicles continue to explore worlds beyond our planet [3].

Regardless of the vehicle's exact usage, navigation is one of the fundamental issues facing the autonomous operation of vehicles - they need to go where they need to be to complete their assigned task [4]. To do so, a vehicle must be capable of identifying a collision free path from one point to another, a problem that can be further broken down into a number of sub-tasks that includes map building, localization, path planning, and obstacle avoidance. Of these various sub-tasks, the ability to detect and react to obstacles in real time is arguably the most crucial to AVs as it ensures the safety of the vehicle itself, people, wildlife and other vehicles it may come in contact during its operation. AVs rely on sensors for perception of its environment, thus the correct selection of sensors which can accurately pick up the obstacles that a vehicle might encounter in its intended usage environment is required.

To this end, light detection and ranging, or LiDAR sensors have emerged as one of the most ubiquitously used sensors in various AVs. These highly accurate laser sensors which are resilient to background light interference are being used for multiple tasks pertaining to machine autonomy, including but not limited to simultaneous localization and mapping (SLAM) [5], obstacle avoidance [6] and even object detection [7]. However, as powerful as they are, no single sensor can function perfectly under all conditions and identify all kinds of obstacles without fail. Sensor fusion is being looked into as the solution as multiple sensors sporting different strengths can increase the collision avoidance capability of an AV.

This paper aims to explore the application of sensor fusion with LiDAR to improve the collision avoidance capabilities of vehicles operating under various environments, as well as some of the reactive control algorithms used to process the data collected and enact avoidance strategies. The framework of this paper is as illustrated in Fig. 1, and the main contributions of this paper are outlined as follows:

- Provides a comprehensive review of recent advancement in the various sensor technologies used for obstacle avoidance in AVs over the last decade, emphasizing on the improvement offered by the fusion of different types of sensors with LiDARs.
- Identifies current research trends in enhancing the obstacle avoidance capability of indoor and outdoor AVs, focused on the choice of sensors for fusion and the motivation behind these choices.
- Discusses the prevailing challenges faced by AVs operating under different environments and suggest future research avenues to improve the obstacle avoidance accuracy and reliability of AVs.

The outline of this paper is as follows: Section II depicts the methodology used to gather data for this review paper. Section III summarizes the principles and types of LiDAR sensors, and Section IV delves into the various types of sensors used in obstacle avoidance as well as the pros and cons when used in fusion with LiDARs. Next, Section V presents the various reactive obstacle avoidance algorithms that are used today, followed by Section VI which tabulates recent publications and analyzes the research trends, challenges and future directions of LiDAR-based obstacle avoidance in AVs. Finally, Section VII provides a conclusion for the paper.

II. METHODOLOGY

Based on the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) 2020 method, we have reviewed a number of recent articles focused on the collision avoidance of LiDAR-based AVs, the outline of which is as shown in Fig. 2.

We initiated our research by conducting searches in two reputable scientific databases, namely Web of Science (WoS) and Scopus, to compile a comprehensive list of publications pertaining to sensor fusion for obstacle avoidance in LiDAR-based AVs. The keywords utilized for this search are detailed in Table 1. We restricted our search to papers published in English between January 2015 and April 2024. The search yielded 406 and 460 records from WoS and Scopus, respectively, totaling 866 records before screening. Additionally, we conducted supplementary searches using platforms like Google Scholar and online tools such as ResearchRabbit, which yielded an additional 30 and 10 records, respectively. Consequently, the total number of records identified in this phase amounted to 906.

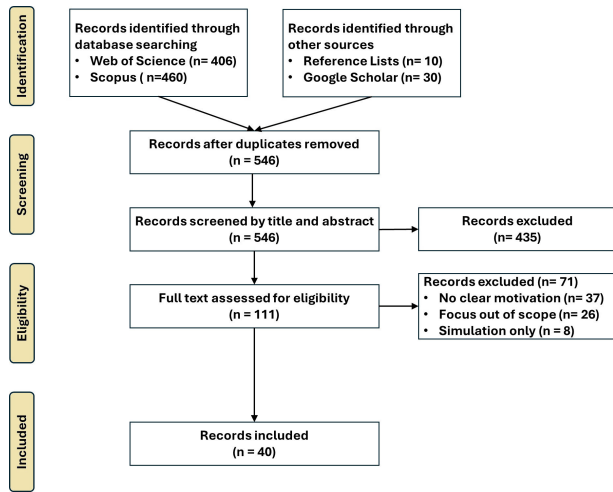
Initially, the 906 records undergo a screening process to identify and remove duplicates, resulting in 546 unique entries. Subsequently, manual screening is conducted without the use of automation tools. A title screening eliminates 220 records, followed by abstract screening. This reduces the pool to 111 records, which are then assessed by reviewing the full text.

The inclusion criteria for this review encompass:

- (i) Description of sensor fusion involving LiDAR and at least

TABLE 1. Keywords and filters used in preliminary database search.

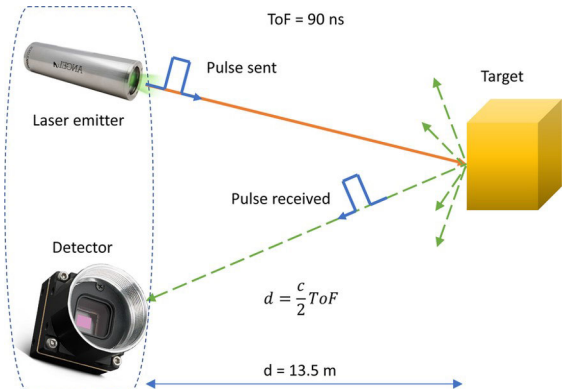
Criteria	Search Terms
Database	Web of Science OR Scopus
Search Field	Title, Keyword and Abstract
Keywords	((collision avoidance) OR (obstacle avoidance) OR (obstacle detection) OR (avoid) OR (detect)) AND ((lidar) OR (laser)) AND ((sensor fusion) OR (multi sensor)) AND ((autonomous mobile robot) OR (autonomous vehicle) OR (unmanned aerial vehicle) OR (unmanned surface vehicle) OR (autonomous car) OR (self-driving car) OR (AMR) OR (UGV) OR (UAV) OR (USV))
Language	English
Publication Date	FROM January 2015 TO April 2024

**FIGURE 2.** Flow diagram depicting the identification and selection of publications to be included in this review.

one other exteroceptive sensor (e.g., vision, ultrasonic, IR, or radar); (ii) Intended application for collision avoidance in AVs; (iii) Explicit mention of the motivation behind sensor selection; (iv) Focus on how sensors inform collision avoidance methods, excluding details on training, data fusion, and calibration; (v) Publication as a journal article or conference proceeding; and (vi) Conducting experiments rather than relying solely on simulations. Conversely, rejection criteria include: (i) Publications deviating from the inclusion criteria; (ii) Exclusion of review papers, surveys, and book chapters; (iii) Lack of clear motivation for sensor selection; (iv) Reliance solely on simulation without experimental validation; and (v) Unavailability or inaccessibility of the full text. After applying the inclusion and exclusion criteria, 40 papers are selected for analysis in this paper.

III. LiDAR PRINCIPLES AND TYPES

LiDARs are active sensors which emits lasers onto a target and measure the time taken for the reflected or backscattered light to return to the detector, a principle called time-of-flight (ToF) as illustrated in Fig. 3. By expanding on this simple principle, a wealth of information can be gleaned from the target, among them its distance, velocity and even chemical composition [8]. While LiDARs have been around since the 1960's - various recent breakthroughs such as scanning

**FIGURE 3.** Illustration of LiDARs' ToF principle [11].

technologies, computational power and pattern recognition algorithms have contributed to the recent boom in LiDAR adoption. Today's LiDAR sensors enjoy high measurement precision, large detection range and fast data acquisition [9], all while its physical size, power consumption and production costs have gone down [10].

The multitude of data, combined with the performance improvement has allowed LiDARs to find purchase in many diverse fields and applications. In general, the information that can be obtained using a LiDAR sensor can be broadly categorized into spatial, spectral and temporal types [10]. Spatial information refers to the measurement of its surroundings like distances, which is most important in mapping and imaging tasks. This is usually achieved by producing a point cloud of measurements that can be interpreted using various image processing algorithms, and are commonly utilized in the autonomous navigation of AVs. Spectral information relates to the material an object is made of, an information that can be obtained by measuring by the amount of light reflected off the target surface. As such, spectral information is mostly used in object identification, for example in agricultural robots. Temporal data, as the name suggests, refers to the repeated measurement of an environment over a set period of time. LiDARs used in remote sensing make use of temporal data, such as in the monitoring of atmospheric constituents.

A LiDAR is composed of two main components, namely a laser rangefinder and a beam deflection or scanning mechanism. The laser rangefinder, sometimes termed as a one-dimensional (1D) LiDAR [12], consists of emitter and receiver circuits, and is responsible for the generation,

transmission and detection of the laser beams to measure the distance of a target point to the rangefinder. A scanning mechanism rotates the laser rangefinder on its axis to obtain the measurement of all or a subset of points on a plane depending on the LiDAR's field of view (FOV) [10]. The FOV of a two-dimensional (2D) LiDAR is limited only to the horizontal plane, while a three-dimensional (3D) LiDAR's field of view involves both the horizontal and vertical planes. 3D LiDARs can be realized by adding a tilting or sweeping mechanism to a 2D LiDAR, or by stacking multiple 2D LiDARs offset by certain angles, so the x-y coordinates of each 2D sensor can also be analyzed in the z-plane [12].

The beam deflection mechanism of LiDAR sensors has come a long way. In the past, it is confined to mechanical methods, such as by rotating a mirror. This results in the large size of early LiDAR sensors, in addition to the introduction of vibrations, which may impact the accuracy of measurements [10]. Now, electromechanical, micro-electromechanical systems (MEMS) and solid state scanners provide various alternatives to mechanical scanning, each providing various pros and cons, explored in the literature in [10].

A number of measuring principles can be used to calculate the difference between the emitted and the detected beams. A pulsed ToF is a straightforward approach in which a beam of intense light is directed at the target over a small fraction of time. The high-energy light pulses minimize the disturbance of background illumination and is considered safe for human eyes. This approach can be used for both indoor and outdoor applications over very long distances, however it has a poor signal-to-noise performance as more emitted light scatters off distant targets [13].

While a pulsed laser source emits light signals in intermittent bursts, a continuous wave laser sources outputs signals continuously over time. Modulation can be applied to continuous wave sources to overcome some of the limitations of pulsed sources. In the amplitude-modulated continuous wave (AMCW) approach, the laser's amplitude is adjusted at a consistent frequency, enabling the detector to capture the phase shift of reflected light. By selecting a higher emitter signal frequency, the measurement precision of this method can be enhanced at the cost of reducing its detection range, making AMCW less suitable for extreme distances. Nonetheless, its widespread availability sustains its use in indoor settings. Conversely, frequency-modulated continuous wave (FMCW) alters the laser's frequency and measures the disparity between emitted and received signals. This method facilitates simultaneous distance and velocity measurements through the Doppler effect. However, FMCW is susceptible to environmental temperature fluctuations, posing a significant drawback.

IV. SENSORS USED IN OBSTACLE AVOIDANCE

A. LiDAR

The selection of obstacle avoidance sensors for autonomous vehicles (AVs) largely hinges on the types of obstacles

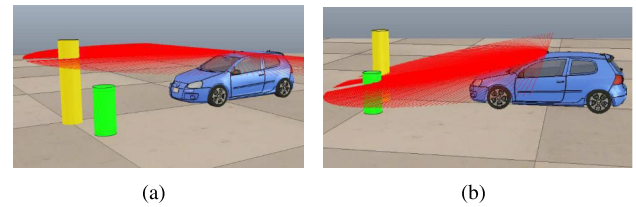


FIGURE 4. Effect of LiDAR scan angle on obstacle detection: a) LiDAR parallel to the ground detects higher obstacles only, while b) inclined LiDAR detects obstacles at various heights [12].

prevalent in their operational environment. These obstacles are typically categorized as positive or negative, with positive obstacles protruding upwards from the ground (e.g., slopes or walls), while negative obstacles represent ground depressions (e.g., holes or cliffs) [14]. While negative obstacles are generally not directly observable by most sensors, they are inferred based on variations in readings from the two closest points to the obstacle.

In recent years, 3D LiDAR scanners have become indispensable for outdoor self-driving vehicles [15]. These scanners produce precise real-time 3D representations of obstacles, enabling vehicles to perceive a broad area and react swiftly to diverse road conditions and obstacles [16]. However, the extensive data points generated by 3D LiDARs pose processing challenges, contributing to increased processing time and power consumption. Additionally, their bulky size and high cost present practical limitations.

For research prototypes and small-scale indoor vehicles, 2D LiDARs are often favored due to their cost-effectiveness, lower energy consumption, and simplified processing requirements. Nevertheless, 2D LiDARs only capture data within a single plane, limiting their effectiveness in scenarios with irregularly shaped objects or varying terrain. Strategies to mitigate these limitations include controlling the scan angle and mounting height of the LiDAR on the vehicle body [17], [18]. Fig. 4 illustrates how adjusting the LiDAR's orientation affects obstacle detection. From the figure, it can be seen that for a 2D LiDAR attached to the top of a vehicle with a scan plane parallel to the ground, only obstacles about the same height or higher than the vehicle will be detected, whereas a downwards scanning plane allows shorter obstacles to be picked up as well.

Numerous mobile robot prototypes for indoor usage relies solely on LiDAR sensors. The literature in [6] has demonstrated that a system based on a single LiDAR sensor is able to accurately measure targets from 0.12 to 10.5 meters, and can avoid obstacles with similar or larger size to the prototype, unaffected by ambient light intensity and the obstacle's colors. However, the sensor fails to detect transparent objects. Meanwhile, in [19] the prototype car can drive between and avoid multiple obstacles much taller than the car, but the author notes that accumulated noise from the LiDAR sensors would need to be improved.

On the other hand, [17] listed a number of situations where 2D LiDAR falls short of the task, such as on a UAV. The

LiDAR sensors are able to detect obstacles but not avoid them entirely as the vehicle operates in a 3D space. Another example given in the same work is the presence of high luminosity target in the scanning direction of the sensor, causing sensor saturation. The author proposes the usage of a rotating mirror with a 2D LiDAR as a low cost solution to obtain a 3D perception. The accuracy of measurement in a 360 degrees field of view is satisfactory at above 96%, although the performance falls off for distant objects. The publication in [20] uses a servo-controlled gimbal to tilt a 2D LiDAR to obtain a 3D map. The prototype is able to create 3D maps autonomously as long as a closed contour of the environment can be detected, with the exception that black-colored surfaces that absorb instead of reflect light appear invisible to the sensor. Another option is to use an inclined laser rangefinder in addition to a 2D LiDAR to detect obstacles below the scanning plane of the 2D LiDAR, as reported in [21].

B. LiDAR AND VISION SENSORS

While LiDAR systems offer excellent depth perception, they often lack object recognition capabilities [22]. Unless equipped with multiple lasers of different frequencies or high beam-count 3D LiDARs with trained object classification models, additional sensors are typically needed for object recognition. Vision sensors, such as cameras, are a natural complement to LiDARs, excelling in object recognition but lacking in depth or distance measurements [23]. Moreover, they are well-researched and cost-effective.

Vision sensors, or cameras, are passive sensors that convert light stimuli into electric signals. They operate in the human visible light spectrum and produce images that can be monochromatic or colorized (RGB). Depth cameras, a subtype, provide distance measurements of objects in addition to graphic information but with higher processing requirements. Extracting useful features from the rich information contained in digital images returned by cameras demands significant computational effort [24]. Additionally, vision sensors are susceptible to interference from background lighting and occlusions, performing poorly in low-light and adverse weather conditions.

By fusing LiDAR and camera data, the location of an obstacle can be accurately measured by laser, while the type of obstacle can be identified via image processing and classifier algorithms, allowing for a more accurate, semantic interpretation of the environment as a whole [25], [26]. There are numerous literature exploring the ways and algorithms to achieve efficient LiDAR-camera fusion, the scope of which is beyond this paper, but generally such a fusion follows the steps of calibration, projection of LiDAR points onto camera image to identify features or regions of interest, and classification of said regions of interests using classifier algorithms, as shown in Fig. 5. Finally, a decision is made based on the result of the classification step [23], [25], [27].

Applications of LiDAR and camera systems include the detection of four-wheeled carts [28]. In this system, 2D



FIGURE 5. General steps of camera-LiDAR sensor fusion involving calibration between the two sensors, projection of LiDAR points onto image for feature extraction, and classification of the identified features.

LiDAR scans detect the legs of carts, while a ToF camera confirms the presence of the cart as the robot approaches the target. Such setups are useful in environments with irregularly shaped objects like carts and tables, particularly indoors. In [25], a 3D LiDAR fused with a monocular camera increases the detection rate of vehicle and pedestrian obstacles in various weather conditions. Additionally, [29] uses a single 3D LiDAR fused with an RGB camera to track human and other obstacles in an industrial production cell, estimating safe interaction speeds between humans and robots in shared spaces.

C. LiDAR AND ULTRASONIC

Ultrasonic sensors, also known as sonars, are widely utilized for collision avoidance applications. These sensors offer several advantages, including affordability, low energy consumption, minimal computational requirements, compact size, and lightweight design. Operating on the ToF principle, ultrasonic sensors emit sound waves and measure the time taken for their echoes to return from obstacles [30]. Typically operating at frequencies above 20 kHz, ultrasonic waves fall outside the audible range of humans. Illustrated in Fig. 6, these waves propagate in a conical beam pattern from the sensor's point source.

Some of the shortcomings of ultrasonic sensors include susceptibility to environmental factors like temperature, humidity and ambient noise. They are also selective about the nature and type of obstacles they can accurately detect: smooth, solid surfaces which reflect sound waves well that are incidental to the transmitter source are optimal, conversely surfaces which absorbs or reflects sound waves away from the source cannot be detected reliably. Ground-based vehicles might run into the problem of an ultrasonic sensor detecting the ground as an obstacle as shown in Fig. 7. The author of [31] recommends placing the ultrasonic sensor higher relative to ground to increase the ground detection distance, but notes that higher sensor placement reduces the probability of detection for smaller obstacles.

Due to the highly similar nature of ultrasonic and LiDAR sensors, they are often fused for use in indoor obstacle detection where the obstacles do not vary too much in nature. Ultrasonic sensors perform reasonably well at short to medium ranges and the data is relatively simple to process, as opposed to the longer ranges and higher complexity of LiDARs. One such example is the AMR described in [32], where The 2D LiDAR used can measure distances from 0.15 to 12m, whereas the ultrasonic sensor

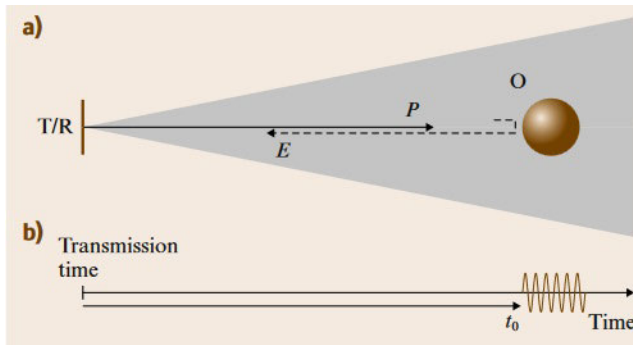


FIGURE 6. Operating principle of pulse-echo ultrasonic sensors: a) T/R denotes the transmitter/receiver, P represents the path taken by the acoustic pulse, E indicates the path taken by the echoes, and O marks the target object. b) shows the corresponding observed waveform on the receiver, with t_0 indicating the transmission time [30].

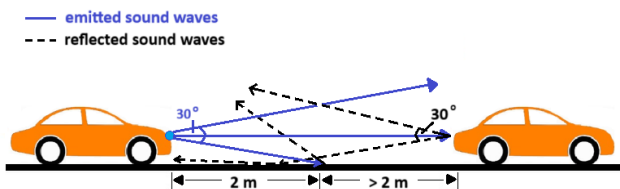


FIGURE 7. Illustration of a false positive detection in an ultrasonic sensor setup: The ground surface reflects sound waves at distances greater than 2 meters, causing the sensor to mistakenly detect the ground as an obstacle, adapted from [31].

can accurately measure distances from 0.02 to 0.4m. The data fusion between both type of sensors increases the obstacle detection range and reduces the errors incurred by each sensor separately, thus improving the overall accuracy of the system. Additionally, in indoor environments, transparent obstacles like glass doors might also be present, which cannot be detected by LiDARs but are not a problem for ultrasonic sensors. Multiple publications have leveraged on a sonar-LiDAR fusion system for the detection of transparent and optically reflective obstacles [33], [34].

In [18], a novel 3D ultrasonic sensor - which utilizes an array of sound signal emitters and receivers to obtain 3D position data of objects - is fused with a 2D LiDAR for the mapping task of a mobile robot. The 3D ultrasonic sensor is capable of detecting obstacles that the LiDAR is unable to, such as objects on the ground and tables higher than the robot, and both the 2D and 3D maps generated through data fusion are more accurate compared to when using only one sensor.

A small quadrotor UAV combining four ultrasonic sensors and one LiDAR sensor for landing and obstacle avoidance is tested in [35]. It is observed that the ultrasonic sensors have better accuracy in distance measurement than LiDARs at ranges below 5m; and while both are susceptible to errors when the environmental temperature is increased by 15 degrees Celsius, the ultrasonic sensors exhibit more distortions. Averaging the readings from both sensors increases robustness and overall accuracy of the system.

D. LiDAR AND RADAR

Radar, originally an acronym for radio detection and ranging, has long been utilized in military and aerospace sectors for object detection and tracking. Similar to LiDAR sensors, radars transmit electromagnetic waves in radio or microwave frequencies and measure the reflected signal off objects, providing accurate readings of speed and range. Despite their similarities, radars were not widely adopted in robotics until recently due to several factors [36]. Legal restrictions on bandwidth, originating from military applications, have historically limited the range resolution of commercial radar sensors, especially in frequency-modulated continuous wave radars [37]. Additionally, radar sensors were once costly, complex, power-hungry, and bulky [38], constraining their use on smaller battery-powered vehicles for indoor applications in the past.

Most radars used in AVs are of the FMCW variety. Compared to pulsed radars, continuous wave radars have shorter detection ranges and higher power consumption due to continuous signal transmission. However, they offer improved resolution and SNR ratio by continuously listening for signals.

Radar signals, with wavelengths typically larger than airborne particles like dust, smoke, and fog, remain unaffected by aerosols, unlike LiDAR signals [37]. Radar data also require less complex algorithms for processing compared to LiDARs [39]. However, radar data are susceptible to noise and require effective filtering, and they offer lower detection ranges and resolutions than LiDARs. Combining these two sensors enhances system performance, particularly in low-visibility conditions. Radars excel in measuring obstacle velocities but are less precise in determining positions, whereas LiDARs exhibit the opposite behavior [40]. Consequently, LiDAR-radar combinations are common in applications involving dynamic obstacles, such as autonomous driving.

For instance, a rescue robot presented in [37] integrates 3D LiDAR and 2D radar data for indoor obstacle detection and simultaneous localization and mapping (SLAM). Radar data supplement LiDAR data, enhancing system robustness in challenging conditions with smoke, dust, or fog. However, the system offers no advantage over typical LiDAR-based systems in clear environments. LiDAR-radar fusion is also employed to eliminate moving objects for SLAM in dynamic indoor and outdoor environments [5], where the presence of moving obstacles undermines map accuracy in traditional LiDAR-based SLAM.

In outdoor obstacle detection for UAVs, [41] explores the fusion of sonar, radar, and LiDAR data. Radars effectively detect obstacles challenging for LiDARs, such as glass windows, poles, and nets, enabling the UAV to identify a broader range of outdoor obstacles. Similarly, [40] demonstrates the fusion of multiple 3D LiDARs and radar for autonomous driving, enhancing measurement accuracy for moving obstacles and minimizing non-detection periods. Lastly, [42] investigates a LiDAR-radar system to detect

occluded pedestrians, where LiDARs identify occlusions like trees and signboards, aiding in detecting pedestrians hidden behind them.

E. LiDAR AND INFRARED

Another type of optical sensor which are used in obstacle avoidance is the infrared (IR) sensor. Instead of the visible light used by cameras and laser used in LiDARs, IR sensors detect electromagnetic waves in the infrared order or heat. IR sensors come in both active or passive varieties, with active IR sensors emitting IR radiation and measuring the reflected light off of objects, while passive IR sensors detect the change in temperature or thermal radiation given off by human or animal targets [43]. The operating principles of both active and passive IR sensors are as illustrated in Fig. 8.

Active IR sensors are highly similar to ultrasonic sensors, in that they are cheap, simple and lightweight solutions for low to mid-range distance measurement, and additionally they have higher response times compared to the latter. However, as they are optical sensors, IR sensors are subject to background light interference, are unable to detect transparent and reflective objects, and if used outdoors, highly susceptible to weather conditions like sunlight, moisture and humidity. Passive IR sensors require even less power and has high accuracy within its narrow detection zone, but is additionally susceptible to background heat sources and cannot distinguish between multiple heat sources with the same temperature range, for example two humans passing directly in front of the sensors.

For most outdoor AVs, IR sensing is applied through the use of passive IR cameras, which make use of a large array of photodetectors, as opposed to the much lower number of detectors used in active IR range finders or IR motion sensors. IR cameras - commonly known as night vision cameras - capture images using IR radiation and can function in low light conditions. The resultant image can then be processed in a similar fashion to vision cameras for object detection. Thermal IR cameras also measure the heat radiated from obstacles and are very reliable in detecting humans and animals, a property highly valuable to autonomous driving.

Although LiDAR readings are satisfactory during the day and under clear weather, thermal IR cameras are shown to have better performance when the surroundings is suddenly darkened or under bad weather conditions where airborne particles like rain, fog or dust are present [44]. As such, multiple works have proposed the fusion of LiDAR and thermal IR camera for autonomous driving. Reference [44] shows that the detection probability using both sensors is higher compared to individual sensors under sudden darkness, foggy, and in presence of small human obstacle, while [45] presents a reliable object detection system under both day and night conditions fusing both sensors.

V. OBSTACLE AVOIDANCE ALGORITHMS

A reliable autonomous navigation system lies at the crux of every AV [46]. Extensive research has been conducted

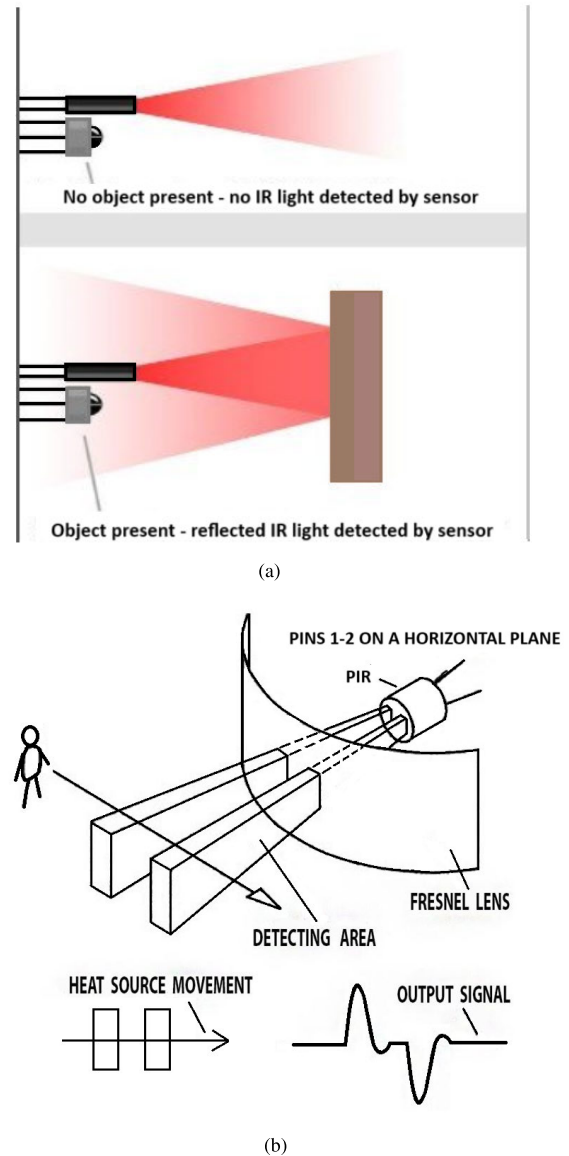


FIGURE 8. Operating principle of a) active IR, which measures the reflected IR light off of obstacles and b) passive IR sensors, which measure the heat radiated from obstacles [43].

on this topic, exploring various control architectures that organize environmental perception into collision avoidance actions [47], [48]. Three common control architectures are deliberative planning, reactive control, and hybrid control, depicted in Fig. 9.

Global path planning, a component of the deliberative planning layer [50], involves planning a trajectory toward a goal, considering path optimization and static obstacles in a world model, as shown by the plan block in Fig. 9(a). This requires an accurate environment representation or map, leading to increased calculations and processing time, which are less favorable when confronted with sudden, dynamic obstacles. In contrast, reactive control, also known as local obstacle avoidance, focuses solely on vehicle safety

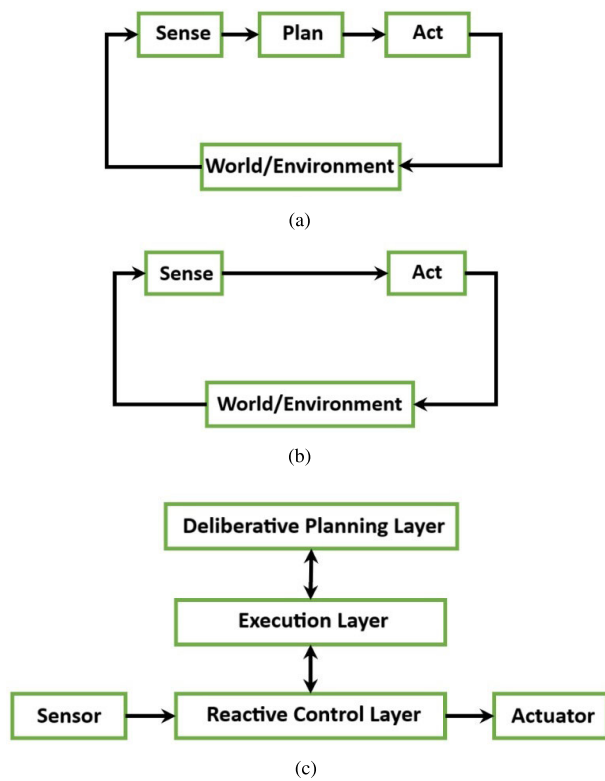


FIGURE 9. Block diagrams of a) deliberative planning, b) reactive control and c) hybrid control architecture [49].

at a low level, considering only the immediate environment and dynamic obstacles [51], [52]. Reactive control directly processes sensor information and reacts accordingly, as depicted in Fig. 9(b), enabling swift responses with minimal computation but potentially diverting the vehicle from its goal or leading it into a local minimum [53]. Ultimately, effective AV operation necessitates both global and local navigation strategies, often requiring a combination of both algorithms [54]. Hybrid control architectures address the weaknesses of both by prioritizing the reactive layer to promptly control vehicle motion in response to potential collisions, while the deliberative layer concurrently finds an updated path to the goal, corresponding to Fig 9(c).

This paper focuses on reactive obstacle avoidance, the lowest level of actuator control akin to reflexes, which directly processes sensor inputs and makes real-time decisions altering the vehicle's motion [55]. As reactive control operates on sensor readings, the algorithm employed must effectively handle uncertain, noisy, and occasionally erroneous data. Two commonly employed algorithms for this task are fuzzy logic and neural networks, both of which are elaborated upon in the subsequent subsections.

A. FUZZY LOGIC

Fuzzy logic is based on the theory that instead of a rigid true-false binary, a logic variable can take on a range of values, not unlike the subjectivity of natural languages that contains

the notion of “true”, “completely true”, “not very true” and so on. In a fuzzy logic system, the inputs activate a set of rules to varying degrees based on a membership function, a step known as fuzzification. A set of IF-THEN rules are then applied, and the output is a combination of all the rules that are activated, which is finally defuzzified to produce a crisp signal used for hardware control [23], [56].

An example of a generic fuzzy controller for obstacle avoidance with two inputs from the sensors S1 and S2, and one output to a motor M1 is as shown in Fig. 10. The distance of obstacles relative to S1 and S2 is obtained and fuzzified into Very Close, Close or Very Far. After applying all applicable rules from the rule base to both input signals, an output that is a member of the sets of Left, Center or Right is produced, and finally this value is defuzzified into an electric signal that is sent to the motor M1.

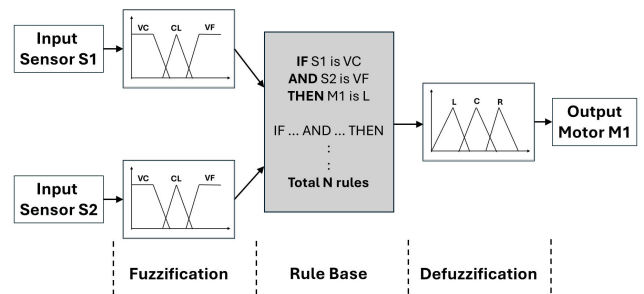


FIGURE 10. An example of a fuzzy controller with two inputs from sensors S1 and S2 and one output to motor M1. Membership functions legends: VC: Very close, CL: Close, VF: Very Far, L: Left, C: Center, R: Right.

Fuzzy systems are inherently similar to human reasoning, it divides the problem into smaller objectives and makes decision based on uncertain and imprecise information. It simplifies the solution, allows for adding of additional rules and objectives without significantly altering the system, and enables an easy modeling of human expertise [57]. Additionally, as sensor readings are not always precise and may contain errors, fuzzy logic is well suited to working with these uncertainties [58]. Although, on the downside, fuzzy logic might not be sufficient to solve problems which require high precision, as the large amount of rules needed to ensure that precision will no doubt slow the system down.

The literature in [59] compares three different fuzzy controllers for a waiter robot, and arrives at the conclusion that more input variables and more defined rules generates faster collision avoidance and smoother trajectory in said robot, but the addition of more inputs caused the number of rules to increase exponentially.

B. NEURAL NETWORKS

Neural networks (NN), also known as artificial neural networks, are machine learning algorithms utilized for tasks such as classification, pattern recognition, and function approximation [60]. These networks emulate the functioning of biological brains, comprising interconnected processing units termed nodes or neurons. When a node's input signal

surpasses a threshold known as the activation function, it generates and transmits an output signal to another node. Input nodes respond to external stimuli like sensor signals, while subsequent nodes activate based on weighted connections from active nodes in preceding stages [61]. NNs are broadly categorized by signal flow direction, with feedforward NNs facilitating signal traversal from input to output nodes and recurrent NNs allowing outputs to connect back to input nodes [62].

Nodes are structured into layers, each applying specific transformations to input signals. A basic NN typically comprises an input layer, at least one hidden layer, and an output layer. The hidden layer is so named because external observers lack insight into its processes [63]. NNs may feature multiple hidden layers, with those containing more than two termed deep neural networks. Greater hidden layers enable the system to solve more complex problems, albeit at the cost of increased processing time. Fig. 11 illustrates a feedforward NN with two hidden layers.

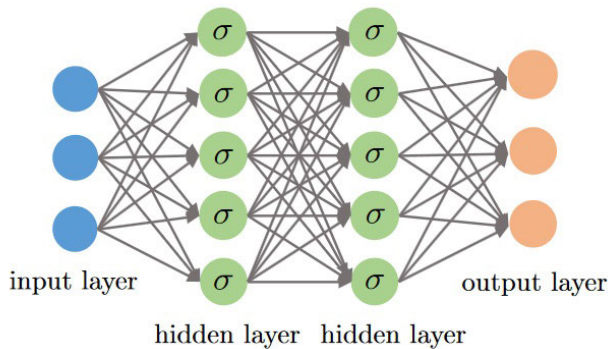


FIGURE 11. An example of a feed-forward neural network comprising of one input layer, two hidden layers and an output layer [64].

The essence of NNs lies in their learning capability. By comparing produced results to desired outputs, NNs adjust node weights to generate new results closer to desired ones. With ample training data, errors diminish, enabling trained NNs to transform input into desired outputs swiftly and accurately. NNs deployed for obstacle avoidance typically accept sensor data as inputs and produce robot motor speed, direction, or angle as outputs [65]. Various systems implementing obstacle avoidance via NNs are documented in prior literature [66]. However, NNs necessitate model training, which can span several days, and demand higher hardware specifications for smooth operation [67].

For this paper's purposes, publications utilizing NNs to classify obstacles without specifying dedicated reactive collision avoidance algorithms are considered as leveraging NNs for obstacle avoidance. This is because navigation commands can be executed based on identified obstacles.

VI. ANALYSIS AND POTENTIAL RESEARCH DIRECTIONS

A. ANALYSIS

A total of 40 publications from the past decade (2015-2024) which describes the fusion of LiDAR and at least one

other sensor for obstacle avoidance in AVs, which include autonomous cars, AMR, USV and UAV, has been reviewed and summarized in Table 2. The publications are sorted by their intended operating environment, whether outdoors or indoors, vehicle type, and LiDAR type, and includes the LiDAR make, secondary sensor type, detection range, sensor fusion objective and obstacle avoidance algorithm.

A majority (54.5%) of the outdoor AVs are autonomous cars (ACs), followed by AMRs (22.7%), with ASVs and UAVs trailing at 13.6% and 9.1%, respectively. This trend aligns with the rapid development of autonomous passenger cars, driven by the need for enhanced safety for passengers and pedestrians. AMRs are also actively being developed, particularly for field and rescue operations or agriculture, while UAVs and USVs remain predominantly within research and academic fields. Research on indoor AVs is overwhelmingly dominated by AMRs (94.7%), with the remainder being a UAV (5.3%). AMRs are well-established in industrial, logistics, and service sectors, necessitating collision avoidance due to their increased presence among human coworkers and the general public. UAVs, while less likely to collide with humans, still need to avoid static obstacles such as walls, ceilings, and other UAVs operating in the same airspace.

3D LiDARs are the preferred sensor for most outdoor vehicles (59.1%), especially ACs, where they are used in 75% of the reported publications. This preference is due to the high safety requirements for on-road cars traveling at high speeds, where inaccurate or delayed obstacle detection could result in life-threatening situations. Consequently, most publications prioritize the 360-degree FOV, quick response time, and high resolution of these state-of-the-art sensors over cost considerations. AMRs and USVs operating in challenging conditions like disaster scenes and water surfaces also require the additional sensing capabilities offered by 3D LiDARs. A smaller number of outdoor ACs and many AMRs opt for 2D LiDARs, likely due to their use in research contexts. In contrast, 78.9% of indoor AVs—all of which are AMRs—use 2D LiDARs, as they do not require the extensive detection range and speed needed for high-speed travel within constrained environments. Interestingly, 1D LiDARs are used in UAVs for both indoor and outdoor applications, likely because they only need to detect obstacles in their direct trajectory.

Regarding the secondary sensor chosen for fusion with LiDAR, Fig. 12 shows that 68.2% of outdoor vehicles use vision sensors. This is essential as outdoor vehicles encounter a wide variety of obstacles, both static (environmental features) and dynamic (pedestrians and other vehicles). Vision sensors enable the classification of these obstacles, providing valuable insights for optimal avoidance strategies. Additionally, 31.8% of these outdoor AVs include radar as a secondary sensor to detect the speed of moving obstacles. Infrared (9.1%) and sonar (4.5%) are less common, likely due to their limited detection range and susceptibility to background interference.

TABLE 2. Summary of a number of recent publications reviewed on AVs which fuses LiDAR and at least one other type of sensor for obstacle avoidance, sorted by operating environment, type of vehicle, LiDAR type and brand, secondary sensor, main objective of the sensor fusion, and obstacle avoidance algorithm used.

Env.	Ref.	Year	Type of Vehicle	LiDAR Type	LiDAR Make	Secondary Sensor				Detection Range (m)	Obstacle Tracking	Challenging Obstacles (High speed, crowded, reflective etc.)	Sensor Fusion Objective					Obstacle Avoidance Algorithm
						Vision	Sonar	Radar	IR				Robustness/Resilience to Background Conditions	Improve Accuracy/Error Reduction	Increase Measurement Range/FOV	Low Cost	Resource Optimization	
Outdoors	[40]	2018	AC	3D	ibeo LUX			✓		0.25	✓	✓		✓				NN
	[45]	2023	AC	3D	Ouster OS-1				✓	0.5			✓					
	[25]	2021	AC	3D		✓							✓					NN
	[68]	2019	AC	3D		✓		✓			✓		✓					
	[69]	2017	AC	3D	Velodyne VLP-16	✓				1		✓	✓					Neuro Fuzzy
	[23]	2018	AC	3D	Quanergy M8	✓				0.5		✓	✓					
	[5]	2021	AC	3D	Velodyne VLP-16			✓		1		✓						NN
	[42]	2017	AC	3D	Velodyne VLP-16			✓		1		✓						
	[70]	2020	AC	3D		✓								✓				NN
	[44]	2019	AC	2D	Hanback				✓	0.15			✓					
	[71]	2020	AC	2D	RPLidar A1	✓				0.15				✓				NN
	[72]	2018	AC	2D	Neato XV-11	✓				0.2			✓					
	[37]	2016	AMR	3D	Velodyne VLP-16			✓		1			✓					NN
	[73]	2020	AMR	2D		✓									✓			
	[74]	2023	AMR	2D	RPLidar S2	✓				0.05						✓		Fuzzy
	[75]	2020	AMR	2D	RPLidar A1	✓				0.1		✓						
	[76] ¹	2020	AMR	2D	Hokuyo UTM-30LX	✓				0.1						✓	✓	NN
	[77]	2018	UAV	2D	Scanse Sweep	✓				0.1						✓	✓	
	[41]	2017	UAV	1D	Lidar Lite		✓	✓		0.1		✓				✓	✓	NN
	[78]	2018	USV	3D	Velodyne HDL-32E	✓		✓		1			✓					
	[79]	2021	USV	3D	Velodyne VLP-16	✓				1			✓					NN
	[80]	2019	USV	3D	Velodyne VLP-16	✓				1				✓				
Total: 22 publications						15	1	7	2		2	7	9	4	1	3	3	
Indoors	[29]	2020	AMR	3D	Quanergy M8	✓				0.5	✓	✓						Neuro Fuzzy
	[32]	2019	AMR	2D	RPLidar A1		✓			0.02				✓				
	[81]	2022	AMR	2D		✓						✓				✓		NN
	[82]	2020	AMR	2D	Hokuyo	✓						✓				✓		
	[83]	2017	AMR	2D		✓					✓	✓				✓		NN
	[84]	2019	AMR	2D			✓					✓				✓		
	[34]	2021	AMR	2D	Robotis LDS-01	✓	✓			0.02		✓				✓		NN
	[18]	2021	AMR	2D	Sick LMS100		✓			0.5		✓			✓			
	[85]	2016	AMR	2D	Hokuyo URG		✓			0.06	✓						✓	NN
	[26]	2021	AMR	2D	RPLidar A3	✓				0.15				✓			✓	
	[86]	2021	AMR	2D	Robotis LDS-01	✓	✓			0.12	✓	✓						NN
	[87]	2018	AMR	2D	Sick LMS151		✓		✓	0.03					✓			
	[88]	2023	AMR	2D	RPLidar A2	✓				0.15				✓				Fuzzy
	[33]	2018	AMR	2D								✓				✓		
	[89]	2021	AMR	2D			✓		✓	0.1						✓	✓	Neuro Fuzzy
	[76] ¹	2020	AMR	2D	Hokuyo UTM-30LX		✓			0.1					✓	✓	✓	
	[90]	2017	AMR	1D	Lidar lite V3				✓	0				✓				Fuzzy
	[91]	2020	AMR	1D	Lidar lite V3				✓	0				✓				
	[35]	2018	UAV	1D	Lidar lite V3				✓	0						✓	✓	Fuzzy
Total: 19 publications						8	12	0	3		4	6	0	5	4	5	7	

AC: Autonomous car, AMR: Autonomous mobile robot, USV: Unmanned surface vehicle, UAV: Unmanned aerial vehicle.
 NN: Neural network.
 Minimum detection distance is taken from 1) data provided by the publication or 2) the shortest value between all known sensors which specs can be found.
 For the analysis of this paper, publications that employs NNs to classify obstacles but do not specify any other dedicated reactive collision avoidance algorithms are treated as using NN for obstacle avoidance.
 [1] Cited work features an indoor/outdoor design and is separately tabulated in both categories. Total number of publications reviewed is one less than the sum of the two subtotals, i.e. 40.

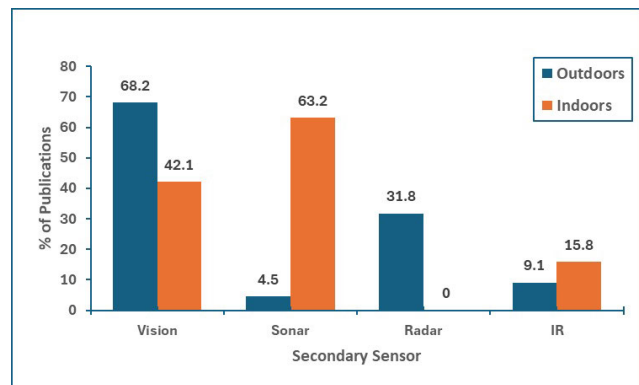


FIGURE 12. No. of publications against types of secondary sensor used for fusion with LiDAR.

A different trend is observed for indoor vehicles. Given the proximity of indoor obstacles to the ego vehicle, low-to-medium range ultrasonic sensors are the preferred choice (63.2%). Vision sensors remain popular at 42.1% for tracking moving obstacles in crowded environments and object detection. Infrared sensors (15.8%) are rarely used and typically accompany another close-range sensor, possibly due

to interference from indoor heat sources. Radars, with their low resolution, are understandably not used at all (0%) for indoor vehicles.

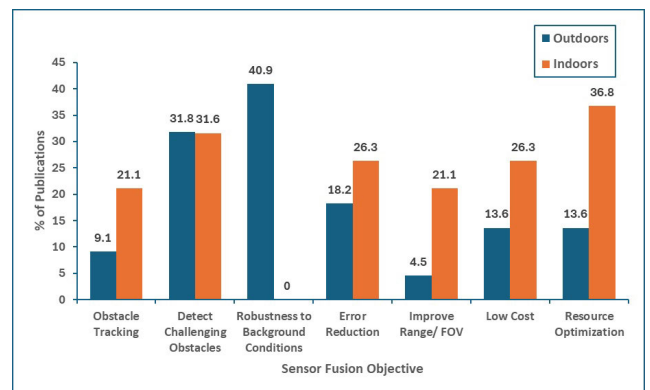


FIGURE 13. No. of publications against sensor fusion objective.

Various reasons are listed by the reviewed publications as the motivation to apply sensor fusion, as illustrated in Fig. 13. For outdoor vehicles, the most direct objective is for the vehicle to safely operate under a variety of

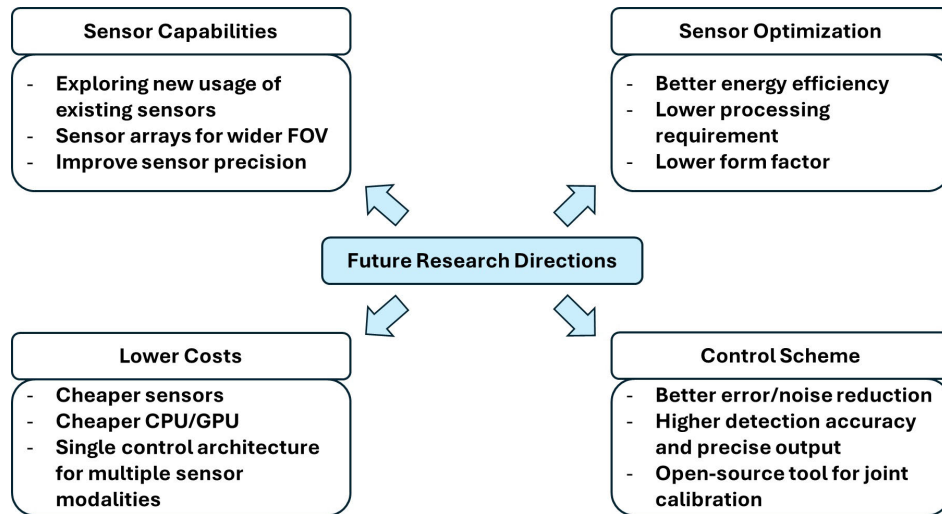


FIGURE 14. Some potential future research directions for LiDAR-based obstacle avoidance in AVs.

conditions (40.9%), be it bad weather like rain, snow or fog, or under certain environmental factors like higher surrounding temperatures. This is closely followed by the need to detect a variety of challenging obstacles (31.8%). The need to improve obstacle detection accuracy (18.2%), using low cost sensors (13.6%), optimize computation or energy resources (13.6%), tracking moving obstacles (9.1%) and improving the measurement range or FOV (4.5%) fill out the rest of the list, each of these factors highly dependent on the purpose of the AV itself. ACs might be concerned with tracking other vehicles' velocities and as such pick sensors that allow them to do so, while UAVs are likelier to pick lighter and energy efficient sensors. Improvement of measurement range or FOV is of the least concern to outdoor vehicles, seeing as many of them are already using sensors with very long detection ranges.

For indoor vehicles, the prioritized objective is optimized resource use, such as lower energy consumption and processing requirements (36.8%). Detecting various difficult obstacles, such as transparent or reflective materials, motivates 31.6% of publications. Reducing measurement error and costs are tied (26.3%), as are obstacle tracking and increasing measurement range or FOV (21.1%). This reflects the prevalent use of 2D LiDARs indoors, necessitating compensation from secondary sensors for better obstacle detection. Service robots and cobots, which operate near humans and crowds, also track obstacles to prevent injuries. Additionally, cost becomes a crucial factor as workplaces often deploy multiple AMRs. Robustness under challenging background conditions is not a concern for indoor vehicles, given the controlled nature of indoor environments.

In summary, robustness in obstacle avoidance—whether under adverse weather or occlusions—and the capability to detect various obstacles, from static road boundaries to dense pedestrian and traffic in urban settings, are critical for outdoor vehicles. Performance considerations outweigh cost and resource factors when selecting sensors for sensor

fusion. Consequently, obstacle avoidance algorithms capable of handling large amounts of uncertain data, such as deep learning NNs and evolutionary algorithms, are essential. These NNs should be trained with sufficient data to achieve quick and accurate classification, enabling AVs to respond instantly to various obstacles to ensure safety.

Indoor AVs require effective obstacle detection against objects made of various materials, like transparent and reflective surfaces, and navigating narrow, constrained spaces with human and robot traffic. Given the small size of most indoor AVs, low-cost, lightweight, and energy-efficient sensors are preferred. Thus, simple algorithms capable of handling uncertain data and producing precise outputs, like fuzzy and neuro-fuzzy algorithms, are suitable.

B. POTENTIAL RESEARCH DIRECTIONS

Based on these trends observed in LiDAR-based obstacle avoidance in AVs, a number of potential research directions can be identified, as summarized in Fig. 14. Firstly, to better detect challenging obstacles, sensor capabilities should be enhanced. New usage for underutilized sensors - for example sonars for outdoor AVs and radars for indoor AVs can potentially be explored, especially now that new technology like 3D ultrasonic positioning and object detection and radar for human detection behind walls are emerging. The limited FOV of cheaper sensors like 2D LiDARs, ultrasonic and IR proximity sensors can possibly be overcome with the use of sensor arrays. The noise and precision issues that accompany the aforementioned cheaper set of sensors can also be combated by better modulation of the transmitted waves, as well as improvement to their transmitter and receiver circuitry.

Next, seeing as resource optimization affects a large number of AVs, sensors can be further optimized to use less resources and occupy less space. Active sensors like sonars, radars, and especially LiDARs are energy intensive; coupled with the processing requirement of LiDAR point

clouds and also the secondary sensor of choice possibly means a quick drain of battery power that AVs run on. Better energy efficiency in both the sensors and algorithms used to process said sensor data would improve battery life across all AVs, even more so on indoor ones. On a similar vein, smaller sensor form factors would allow smaller AVs like mobile robots and aerial vehicles to access them. 3D LiDARs and radars were too large and heavy for aerial vehicles up until their recent miniaturization, and further reduction in these sensors' form factors would both allow for more sensors to be used and free up space and weight on the vehicle.

The cost of the obstacle avoidance system is another area that can be improved upon. Sensors undoubtedly make up a huge portion of the costs, and among the various sensors discussed in this work, 3D LiDARs still tops the chart when it comes to price, which in turn limits their use in smaller scale projects. LiDAR technology as a whole has come a long way in terms of manufacturing costs, so it is likely for 3D LiDARs to undergo a similar trend. An improvement in processing algorithm as discussed earlier would also translate to cheaper hardware like CPUs and GPUs, which would further bring down the cost of the overall system. One of the possible ways to do this is to reuse the control algorithm for multiple sensors, for example instead of separate NNs being used for the vision sensor and LiDARs, a single NN is used instead.

Last but not least, research can be done on the collision avoidance control schemes to better filter and reduce noise and errors from the sensor data, thereby achieving a higher detection rate. A more precise control over the outputs, like smaller changes in the angle and heading of the AV can also be done for a smoother obstacle avoidance path without abrupt brakes and turns. The use of multiple sensors also requires some form of calibration, and an open-source calibration tool that works with multimodal sensors and is easily available to both researchers and vendors would greatly benefit future research on sensor fusion for obstacle avoidance.

VII. CONCLUSION

In conclusion, as AVs take on more tedious and dangerous tasks, effective obstacle detection and avoidance remain crucial for their successful operation. LiDAR sensors are widely favored for their excellent measurement range and resilience under various conditions, yet they have notable limitations, such as difficulty detecting transparent objects and functioning in the presence of aerosols.

To address these limitations, sensor fusion combines data from multiple types of sensors, including vision, ultrasonic, radar, and infrared, to enhance obstacle detection capabilities. This review of 40 recent publications highlights the benefits of sensor fusion in improving system robustness, detecting challenging obstacles, optimizing cost and resource consumption, measuring multiple physical properties, reducing detection errors, and increasing the range and FoV. For outdoor AVs, systems must be resilient to background interference and capable of detecting diverse obstacles

quickly, whether they are high-speed vehicles, dense crowds, or unexpected road conditions. Indoor AVs, on the other hand, need to detect irregularly shaped objects and navigate human and robot traffic efficiently while minimizing energy, weight, and size constraints.

Ultimately, enhancing sensor fusion systems to detect diverse obstacles remains a top priority. Future research should focus on improving sensor capabilities, optimizing resources, reducing costs, and refining control schemes for higher accuracy and precision. This includes expanding the FoV, reducing noise and errors, and developing effective calibration tools for multiple sensor types. These advancements will contribute to more reliable and efficient obstacle avoidance in AVs, ensuring their safe and effective operation across various environments.

ACKNOWLEDGMENT

The authors would like to acknowledge the feedback from Her Ji Ng and Siang Kok Chia, for their valuable insights regarding challenges faced in the manufacturing lines at SanDisk Storage Malaysia Sdn. Bhd.

CONFLICT OF INTEREST

The authors declare no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

REFERENCES

- [1] A. Faisal, T. Yigitcanlar, M. Kamruzzaman, and A. Paz, "Mapping two decades of autonomous vehicle research: A systematic scientometric analysis," *J. Urban Technol.*, vol. 28, nos. 3–4, pp. 45–74, Oct. 2021.
- [2] D. McAslan, M. Gabriele, and T. R. Miller, "Planning and policy directions for autonomous vehicles in metropolitan planning organizations (MPOs) in the United States," *J. Urban Technol.*, vol. 28, nos. 3–4, pp. 175–201, Oct. 2021.
- [3] F. Duarte and C. Ratti, "The impact of autonomous vehicles on cities: A review," *J. Urban Technol.*, vol. 25, no. 4, pp. 3–18, Oct. 2018.
- [4] B. Padmaja, C. V. K. N. S. N. Moorthy, N. Venkateswarulu, and M. M. Bala, "Exploration of issues, challenges and latest developments in autonomous cars," *J. Big Data*, vol. 10, no. 1, pp. 1–24, May 2023.
- [5] X. Dang, Z. Rong, and X. Liang, "Sensor fusion-based approach to eliminating moving objects for SLAM in dynamic environments," *Sensors*, vol. 21, no. 1, p. 230, Jan. 2021.
- [6] D. Hutabarat, M. Rivai, D. Purwanto, and H. Hutomo, "Lidar-based obstacle avoidance for the autonomous mobile robot," in *Proc. 12th Int. Conf. Inf. Commun. Technol. Syst. (ICTS)*, Jul. 2019, pp. 197–202.
- [7] Y. Wu, Y. Wang, S. Zhang, and H. Ogai, "Deep 3D object detection networks using LiDAR data: A review," *IEEE Sensors J.*, vol. 21, no. 2, pp. 1152–1171, Jan. 2021.
- [8] M. Simpson and D. Hutchinson, "Imaging | LiDAR," in *Encyclopedia of Modern Optics*, R. D. Guenther, Ed., Oxford, U.K.: Elsevier, 2005, pp. 169–178.
- [9] Y. Peng, D. Qu, Y. Zhong, S. Xie, J. Luo, and J. Gu, "The obstacle detection and obstacle avoidance algorithm based on 2-D LiDAR," in *Proc. IEEE Int. Conf. Inf. Autom.*, Aug. 2015, pp. 1648–1653.
- [10] T. Raj, F. H. Hashim, A. B. Huddin, M. F. Ibrahim, and A. Hussain, "A survey on LiDAR scanning mechanisms," *Electronics*, vol. 9, no. 5, p. 741, Apr. 2020.
- [11] A. Boschetto, L. Bottini, and L. Macera, "Design and fabrication by selective laser melting of a LiDAR reflective unit using metal matrix composite material," *Int. J. Adv. Manuf. Technol.*, vol. 126, nos. 1–2, pp. 857–872, May 2023.
- [12] O. Yalcin, A. Sayar, O. F. Arar, S. Akpinar, and S. Kosunalp, "Approaches of road boundary and obstacle detection using LiDAR," *IFAC Proc. Volumes*, vol. 46, no. 25, pp. 211–215, 2013.

- [13] S. Royo and M. Ballesta-García, "An overview of LiDAR imaging systems for autonomous vehicles," *Appl. Sci.*, vol. 9, no. 19, p. 4093, Sep. 2019.
- [14] R. D. Morton and E. Olson, "Positive and negative obstacle detection using the HLD classifier," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst.*, Sep. 2011, pp. 1579–1584.
- [15] L. Chen, J. Yang, and H. Kong, "LiDAR-histogram for fast road and obstacle detection," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2017, pp. 1343–1348.
- [16] H. Surmann, A. Nüchter, and J. Hertzberg, "An autonomous mobile robot with a 3D laser range finder for 3D exploration and digitalization of indoor environments," *Robot. Auto. Syst.*, vol. 45, nos. 3–4, pp. 181–198, Dec. 2003.
- [17] V. K. Sarker, L. Qingqing, and T. Westerlund, "3D perception with low-cost 2D LiDAR and edge computing for enhanced obstacle detection," in *Proc. IEEE Conf. Ind. Cyberphys. Syst. (ICPS)*, vol. 1, Jun. 2020, pp. 49–54.
- [18] T. Q. Tran, A. Becker, and D. Grzechca, "Environment mapping using sensor fusion of 2D laser scanner and 3D ultrasonic sensor for a real mobile robot," *Sensors*, vol. 21, no. 9, p. 3184, May 2021.
- [19] Y. Wang, A. Goila, R. Shetty, M. Heydari, A. Desai, and H. Yang, "Obstacle avoidance strategy and implementation for unmanned ground vehicle using LiDAR," *SAE Int. J. Commercial Vehicles*, vol. 10, no. 1, pp. 50–55, Mar. 2017.
- [20] M. G. Ocampo, N. Certad, S. Alvarado, and Á. Terrones, "Autonomous 2D SLAM and 3D mapping of an environment using a single 2D LiDAR and ROS," in *Proc. Latin Amer. Robot. Symp. (LARS) Brazilian Symp. Robot. (SBR)*, Nov. 2017, pp. 1–6.
- [21] P. D. Ren Yee, N. Pinrath, and N. Matsuhira, "Autonomous mobile robot navigation using 2D LiDAR and inclined laser rangefinder to avoid a lower object," in *Proc. 59th Annu. Conf. Soc. Instrum. Control Engineers Jpn. (SICE)*, Sep. 2020, pp. 1404–1409.
- [22] Y. Li and J. Ibanez-Guzman, "Lidar for autonomous driving: The principles, challenges, and trends for automotive LiDAR and perception systems," *IEEE Signal Process. Mag.*, vol. 37, no. 4, pp. 50–61, Jul. 2020.
- [23] P. Wei, L. Cagle, T. Reza, J. Ball, and J. Gafford, "LiDAR and camera detection fusion in a real-time industrial multi-sensor collision avoidance system," *Electronics*, vol. 7, no. 6, p. 84, May 2018.
- [24] J. N. Yasin, S. A. S. Mohamed, M.-H. Hagbayan, J. Heikkonen, H. Tenhunen, and J. Plosila, "Unmanned aerial vehicles (UAVs): Collision avoidance systems and approaches," *IEEE Access*, vol. 8, pp. 105139–105155, 2020.
- [25] B. Sun, W. Li, H. Liu, J. Yan, S. Gao, and P. Feng, "Obstacle detection of intelligent vehicle based on fusion of LiDAR and machine vision," *Eng. Lett.*, vol. 29, no. 2, pp. 1–9, 2021.
- [26] J. Li and R. Stevenson, "2D LiDAR and camera fusion using motion cues for indoor layout estimation," in *Proc. IEEE 24th Int. Conf. Inf. Fusion (FUSION)*, Nov. 2021, pp. 1–6.
- [27] M. Mahlich, R. Schweiger, W. Ritter, and K. Dietmayer, "Sensorfusion using spatio-temporal aligned video and LiDAR for improved vehicle detection," in *Proc. IEEE Intell. Vehicles Symp.*, 2006, pp. 424–429.
- [28] S. Buck, R. Hanten, K. Bohlmann, and A. Zell, "Multi-sensor payload detection and acquisition for truck-trailer AGVs," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, May 2017, pp. 718–723.
- [29] A. Rashid, K. Peesapati, M. Bdiwi, S. Krusche, W. Hardt, and M. Putz, "Local and global sensors for collision avoidance," in *Proc. IEEE Int. Conf. Multisensor Fusion Integr. Intell. Syst. (MFI)*, Sep. 2020, pp. 354–359.
- [30] L. Kleeman and R. Kuc, "Sonar sensing," in *Springer Handbook of Robotics*. Berlin, Germany: Springer, 2016, pp. 753–782.
- [31] J. I. Obuhuma, H. O. Okoyo, and S. O. McOyowo, "Shortcomings of ultrasonic obstacle detection for vehicle driver assistance and profiling," *Int. J. Inf. Technol. Comput. Sci.*, vol. 11, no. 6, pp. 28–36, Jun. 2019.
- [32] S. Premnath, S. Mukund, K. Sivasankaran, R. Sidaarth, and S. Adarsh, "Design of an autonomous mobile robot based on the sensor data fusion of LiDAR 360, ultrasonic sensor and wheel speed encoder," in *Proc. 9th Int. Conf. Adv. Comput. Commun. (ICACC)*, Nov. 2019, pp. 62–65.
- [33] H. Wei, X.-E. Li, Y. Shi, B. You, and Y. Xu, "Multi-sensor fusion glass detection for robot navigation and mapping," in *Proc. WRC Symp. Adv. Robot. Autom. (WRC SARA)*, Aug. 2018, pp. 184–188.
- [34] A. J. Barreto-Cubero, A. Gómez-Espinosa, J. A. Escobedo Cabello, E. Cuan-Urquiza, and S. R. Cruz-Ramírez, "Sensor data fusion for a mobile robot using neural networks," *Sensors*, vol. 22, no. 1, p. 305, Dec. 2021.
- [35] U. Papa, G. Ariante, and G. Del Core, "UAS aided landing and obstacle detection through LiDAR-sonar data," in *Proc. 5th IEEE Int. Workshop Metrol. Aerosp. (MetroAeroSpace)*, Jun. 2018, pp. 478–483.
- [36] I. Arrouch, N. S. Ahmad, P. Goh, and J. Mohamad-Saleh, "Close proximity time-to-collision prediction for autonomous robot navigation: An exponential GPR approach," *Alexandria Eng. J.*, vol. 61, no. 12, pp. 11171–11183, Dec. 2022.
- [37] P. Fritsche, S. Kueppers, G. Briesse, and B. Wagner, "Radar and LiDAR sensorfusion in low visibility environments," in *Proc. 13th Int. Conf. Informat. Control, Autom. Robot.*, 2016, pp. 30–36.
- [38] N. Wessendorp, R. Dinaux, J. Dupeyroux, and G. C. H. E. de Croon, "Obstacle avoidance onboard MAVs using a FMCW radar," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, Sep. 2021, pp. 117–122.
- [39] N. Techaphangam and M. Wongsaisuwat, "Obstacle avoidance using mmWave radar imaging system," in *Proc. 17th Int. Conf. Electr. Eng./Electron., Comput., Telecommun. Inf. Technol. (ECTI-CON)*, Jun. 2020, pp. 466–469.
- [40] H. Hajri and M.-C. Rahal, "Real time LiDAR and radar high-level fusion for obstacle detection and tracking with evaluation on a ground truth," 2018, *arXiv:1807.11264*.
- [41] C. Gianni, M. Balsi, S. Esposito, and P. Fallavollita, "Obstacle detection system involving fusion of multiple sensor technologies," *Int. Arch. Photogramm., Remote Sens. Spatial Inf. Sci.*, vols. XLII-2/W6, pp. 127–134, Aug. 2017.
- [42] S. K. Kwon, E. Hyun, J.-H. Lee, J. Lee, and S. H. Son, "Detection scheme for a partially occluded pedestrian based on occluded depth in LiDAR-radar sensor fusion," *Opt. Eng.*, vol. 56, no. 11, 2017, Art. no. 113112.
- [43] R. Ismail, Z. Omar, and S. Suaibun, "Obstacle-avoiding robot with IR and PIR motion sensors," *IOP Conf. Ser., Mater. Sci. Eng.*, vol. 152, Oct. 2016, Art. no. 012064.
- [44] M.-G. Cho, "A study on the obstacle recognition for autonomous driving RC car using LiDAR and thermal infrared camera," in *Proc. 11th Int. Conf. Ubiquitous Future Netw. (ICUFN)*, Jul. 2019, pp. 544–546.
- [45] J. D. Choi and M. Y. Kim, "A sensor fusion system with thermal infrared camera and LiDAR for autonomous vehicles and deep learning based object detection," *ICT Exp.*, vol. 9, no. 2, pp. 222–227, Apr. 2023.
- [46] A. Loganathan and N. S. Ahmad, "A systematic review on recent advances in autonomous mobile robot navigation," *Eng. Sci. Technol., Int. J.*, vol. 40, Apr. 2023, Art. no. 101343.
- [47] G. Fragapane, R. de Koster, F. Sgarbossa, and J. O. Strandhagen, "Planning and control of autonomous mobile robots for intralogistics: Literature review and research agenda," *Eur. J. Oper. Res.*, vol. 294, no. 2, pp. 405–426, Oct. 2021.
- [48] N. S. Ahmad, "Modeling and hybrid PSO-WOA-based intelligent PID and state-feedback control for ball and beam systems," *IEEE Access*, vol. 11, pp. 137866–137880, 2023.
- [49] S. Ahmed, B. Qiu, F. Ahmad, C.-W. Kong, and H. Xin, "A state-of-the-art analysis of obstacle avoidance methods from the perspective of an agricultural sprayer UAV's operation scenario," *Agronomy*, vol. 11, no. 6, p. 1069, May 2021.
- [50] H. T. Najm, N. S. Ahmad, and A. S. Al-Araji, "Enhanced path planning algorithm via hybrid WOA-PSO for differential wheeled mobile robots," *Syst. Sci. Control Eng.*, vol. 12, no. 1, Dec. 2024, Art. no. 2334301.
- [51] A. Medina-Santiago, L. A. Morales-Rosales, C. A. Hernández-Gracidas, I. Algreto-Badillo, A. D. Pano-Azucena, and J. A. Orozco Torres, "Reactive obstacle-avoidance systems for wheeled mobile robots based on artificial intelligence," *Appl. Sci.*, vol. 11, no. 14, p. 6468, 2021.
- [52] A. Loganathan and N. S. Ahmad, "A hybrid HHO-AVOA for path planning of a differential wheeled mobile robot in static and dynamic environments," *IEEE Access*, vol. 12, pp. 25967–25979, 2024.
- [53] S. Huang, R. S. H. Teo, and K. K. Tan, "Collision avoidance of multi unmanned aerial vehicles: A review," *Annu. Rev. Control*, vol. 48, pp. 147–164, Jan. 2019.
- [54] P. G. Zavlangas, S. G. Tzafestas, and K. Althoefer, "Fuzzy obstacle avoidance and navigation for omnidirectional mobile robots," in *Proc. Eur. Symp. Intell. Techn.*, Aachen, Germany, 2000, pp. 375–382.
- [55] P. Lee, G. Theotokatos, and E. Boulougouris, "Robust decision-making for the reactive collision avoidance of autonomous ships against various perception sensor noise levels," *J. Mar. Sci. Eng.*, vol. 12, no. 4, p. 557, Mar. 2024.
- [56] J. W. Axter and J. R. Bumby, "Fuzzy logic guidance and obstacle avoidance algorithms for autonomous vehicle control," *IFAC Proc. Volumes*, vol. 26, no. 1, pp. 259–264, Apr. 1993.

- [57] A. Fatmi, A. A. Yahmadi, L. Khriji, and N. Masmoudi, "A fuzzy logic based navigation of a mobile robot," *World Academy Sci., Eng. Technol.*, vol. 22, pp. 169–174, Jan. 2006.
- [58] P. Mohanty and D. Parhi, "Controlling the motion of an autonomous mobile robot using various techniques: A review," *J. Adv. Mech. Eng.*, vol. 1, no. 1, pp. 24–39, 2013.
- [59] J. Cerbaro, D. Martinelli, A. S. de Oliveira, and J. A. Fabro, "WaiterBot: Comparison of fuzzy logic approaches for obstacle avoidance in dynamic unmapped environments using a laser scanning system (LiDAR)," in *Proc. Latin Amer. Robot. Symp. (LARS), Brazilian Symp. Robot. (SBR) Workshop Robot. Edu. (WRE)*, Nov. 2020, pp. 1–6.
- [60] N. Syazreen Ahmad, J. Hui Teo, and P. Goh, "Gaussian process for a single-channel EEG decoder with inconspicuous stimuli and eyeblinks," *Comput., Mater. Continua*, vol. 73, no. 1, pp. 611–628, 2022.
- [61] J. Schmidhuber, "Deep learning in neural networks: An overview," *Neural Netw.*, vol. 61, pp. 85–117, Jan. 2015.
- [62] C. H. Goay, P. Goh, N. S. Ahmad, and M. Ain, "Eye-height/width prediction using artificial neural networks from s-parameters with vector fitting," *J. Eng. Sci. Technol.*, vol. 13, pp. 625–639, Mar. 2018.
- [63] S.-H. Han, K. W. Kim, S. Kim, and Y. C. Youn, "Artificial neural network: Understanding the basic concepts without mathematics," *Dementia Neurocognitive Disorders*, vol. 17, no. 3, p. 83, 2018.
- [64] J. Fan, C. Ma, and Y. Zhong, "A selective overview of deep learning," *Stat. Sci.*, vol. 36, no. 2, p. 264, May 2021.
- [65] A.-M. Zou, Z.-G. Hou, S.-Y. Fu, and M. Tan, "Neural networks for mobile robot navigation: A survey," in *Proc. 3rd Int. Symp. Neural Netw.*, Chengdu, China. Cham, Switzerland: Springer, May 2006, pp. 1218–1226.
- [66] M. C. De Simone, Z. B. Rivera, and D. Guida, "Obstacle avoidance system for unmanned ground vehicles by using ultrasonic sensors," *Machines*, vol. 6, no. 2, p. 18, Apr. 2018.
- [67] N. S. Verbitsky, E. V. Chepin, and A. A. Gridnev, "Experimental studies of a convolutional neural network for application in the navigation system of a mobile robot," *Proc. Comput. Sci.*, vol. 145, pp. 611–616, Jan. 2018.
- [68] B. Shahian Jahromi, T. Tulabandhula, and S. Cetin, "Real-time hybrid multi-sensor fusion framework for perception in autonomous vehicles," *Sensors*, vol. 19, no. 20, p. 4357, Oct. 2019.
- [69] T.-E. Wu, C.-C. Tsai, and J.-I. Guo, "LiDAR/camera sensor fusion technology for pedestrian detection," in *Proc. Asia-Pacific Signal Inf. Process. Assoc. Annu. Summit Conf. (APSIPA ASC)*, Dec. 2017, pp. 1675–1678.
- [70] N. Sakic, M. Krunic, S. Stevic, and M. Dragojevic, "Camera-LiDAR object detection and distance estimation with application in collision avoidance system," in *Proc. IEEE 10th Int. Conf. Consum. Electron. (ICCE)*, Nov. 2020, pp. 1–6.
- [71] A. Mulyanto, R. I. Borman, P. Prasetyawana, and A. Sumarudin, "2D LiDAR and camera fusion for object detection and object distance measurement of adas using robotic operating system (ROS)," *Int. J. Informat. Vis.*, vol. 4, no. 4, pp. 231–236, 2020.
- [72] C. Zhou, F. Li, W. Cao, C. Wang, and Y. Wu, "Design and implementation of a novel obstacle avoidance scheme based on combination of CNN-based deep learning method and LiDAR-based image processing approach," *J. Intell. Fuzzy Syst.*, vol. 35, no. 2, pp. 1695–1705, Aug. 2018.
- [73] T. Wang and X. Guan, "Research on obstacle avoidance of mobile robot based on multi-sensor fusion," in *Cyber Security Intelligence and Analytics*. Cham, Switzerland: Springer, 2020, pp. 760–770.
- [74] C. Han, W. Wu, X. Luo, and J. Li, "Visual navigation and obstacle avoidance control for agricultural robots via LiDAR and camera," *Remote Sens.*, vol. 15, no. 22, p. 5402, Nov. 2023.
- [75] Z. Su, C. Wang, Y. Li, and X. Wu, "A method of cliff detection in robot navigation based on multi-sensor," in *Proc. IEEE Int. Conf. Real-Time Comput. Robot. (RCAR)*, Sep. 2020, pp. 152–157.
- [76] M. A. Post, A. Bianco, and X. T. Yan, "Autonomous navigation with open software platform for field robots," in *Proc. Int. Conf. Inform. Control, Automat. Robot.*, Madrid, Spain. Cham, Switzerland: Springer, 2017, pp. 425–450.
- [77] M. Ki, J. Cha, and H. Lyu, "Detect and avoid system based on multi sensor fusion for UAV," in *Proc. Int. Conf. Inf. Commun. Technol. Conver. (ICTC)*, Oct. 2018, pp. 1107–1109.
- [78] L. Stanislas and M. Dunbabin, "Multimodal sensor fusion for robust obstacle detection and classification in the maritime RobotX challenge," *IEEE J. Ocean. Eng.*, vol. 44, no. 2, pp. 343–351, Apr. 2019.
- [79] W. Zhang, F. Jiang, C.-F. Yang, Z.-P. Wang, and T.-J. Zhao, "Research on unmanned surface vehicles environment perception based on the fusion of vision and LiDAR," *IEEE Access*, vol. 9, pp. 63107–63121, 2021.
- [80] M. Sorial, I. Mouawad, E. Simetti, F. Odone, and G. Casalino, "Towards a real time obstacle detection system for unmanned surface vehicles," in *Proc. OCEANS MTS/IEEE SEATTLE*, Oct. 2019, pp. 1–8.
- [81] Y. Han, I. H. Zhan, W. Zhao, J. Pan, Z. Zhang, Y. Wang, and Y.-J. Liu, "Deep reinforcement learning for robot collision avoidance with self-state-attention and sensor fusion," *IEEE Robot. Autom. Lett.*, vol. 7, no. 3, pp. 6886–6893, Jul. 2022.
- [82] J. Liang, U. Patel, A. Jagan Sathyamoorthy, and D. Manocha, "Realtime collision avoidance for mobile robots in dense crowds using implicit multi-sensor fusion and deep reinforcement learning," 2020, *arXiv:2004.03089*.
- [83] T. Dieterle, F. Particke, L. Patino-Studencki, and J. Thielecke, "Sensor data fusion of LiDAR with stereo RGB-D camera for object tracking," in *Proc. IEEE Sensors*, Mar. 2017, pp. 1–3.
- [84] Y. Zhang, X. Wei, and X. Zhou, "Dynamic obstacle avoidance based on multi-sensor fusion and Q-learning algorithm," in *Proc. IEEE 3rd Inf. Technol., Netw., Electron. Autom. Control Conf. (ITNEC)*, Mar. 2019, pp. 1569–1573.
- [85] G. Ciubuciu, A. Filipescu, A. Filipescu, S. Filipescu, and B. Dumitrascu, "Control and obstacle avoidance of a WMR, based on sliding-mode, ultrasounds and laser," in *Proc. 12th IEEE Int. Conf. Control Autom. (ICCA)*, Jun. 2016, pp. 779–784.
- [86] K.-H. Chen, Y.-T. Hwang, and C.-P. Fan, "Sensor data fusion of intelligent autonomous mover for object detection and collision avoidance in environments with surrounding crowds," in *Proc. IEEE Int. Conf. Consum. Electron. (ICCE)*, Jan. 2021, pp. 1–4.
- [87] G. Csaba, L. Somlyai, and Z. Vámosy, "Mobil robot navigation using 2D LiDAR," in *Proc. IEEE 16th World Symp. Appl. Mach. Intell. Informat. (SAMI)*, Feb. 2018, pp. 143–148.
- [88] H. Qiu, W. Chen, A. Ji, and K. Hu, "Research on unmanned vehicle obstacle avoidance technology based on LiDAR and depth camera fusion," *Appl. Math. Nonlinear Sci.*, vol. 9, no. 1, pp. 1–19, Jan. 2024.
- [89] X.-H. Nguyen, V.-H. Nguyen, T.-T. Ngo, and D.-A. Nguyen, "Improvement of control algorithm for mobile robot using multi-layer sensor fusion," *Vietnam J. Sci. Technol.*, vol. 59, no. 1, pp. 110–119, Jan. 2021.
- [90] A. D. Adhvaryu, S. Adarsh, and K. I. Ramchandran, "Design of fuzzy based intelligent controller for autonomous mobile robot navigation," in *Proc. Int. Conf. Adv. Comput., Commun. Informat. (ICACCI)*, Sep. 2017, pp. 841–846.
- [91] S. Adarsh and K. I. Ramchandran, "Neuro-fuzzy based fusion of LiDAR and ultrasonic sensors to minimize error in range estimation for the navigation of mobile robots," *Intell. Decis. Technol.*, vol. 14, no. 2, pp. 259–267, Jul. 2020.



PUI YEE LEONG received the B.S. degree in electrical engineering from the University of Malaya, Malaysia, in 2017. She is currently pursuing the M.Sc. degree in electrical and electronics with Universiti Sains Malaysia, Penang, Malaysia.

From 2018 to 2023, she was a Software Engineer with Intel Microelectronics, Penang, focused on the development of in-house applications for the automation of design verification. Her research interests include automation of the mundane and the application of robotics in high-risk tasks.



NUR SYAZREEN AHMAD (Member, IEEE) received the B.Eng. degree in electrical and electronic engineering and the Ph.D. degree in control systems from The University of Manchester, U.K. She is currently an Associate Professor with the School of Electrical and Electronic Engineering, Universiti Sains Malaysia (USM), specializing in intelligent control, sensor networks, and mobile robotics. Her current research interests include autonomous systems, emphasizing intelligent optimization strategies for motion control, multi-modal sensing, indoor navigation, and human–robot interaction.

...